moment to reflect on a certain bespectacled billionaire who is made fun of for a similar statement made more than twenty years ago.

With 64-bit addresses, processors can address 2^{64} memory locations, or 16 *exabytes* of memory. That's 16 billion gigabytes. To put things in a little perspective, a study by the University of Berkeley in 2003 estimated the size of the Internet at 500 million gigabytes. Even if that number has doubled by today, a 64-bit processor could theoretically load *all the Internet* into memory sixteen times over!—assuming, of course, that it has that much memory available to it. On a smaller scale, imagine being able to load a DVD movie into memory—you'd never have to deal with a skipped frame ever again! Or a time when *Half-Life 2*'s annoying "Loading" message is completely non-existent.

What Is It Good For?

Applications that benefit the most from the move to 64-bit computing are those that involve huge amounts of data and/or calculations. The capability to address embarrassingly large amounts of memory isn't the only advantage that a 64-bit processor has—it also has 64-bit registers (which are used as short-term storage, even before the processor's cache memory). These registers are usually used to store numbers that the processor will use in calculations. Doubling the bits in a number results in the ability to represent it with more precision—for example, pi can be represented correctly to more decimal places using 64 bits. While it is possible for 32-bit processors to work with 64-bit numbers, they need to use two 32-bit registers, and take twice as long to process them. The back-of-the-book version is that 64-bit processors can crunch larger numbers, faster. Audio/Video encoders, financial applications, games and database servers, rejoice! And we're not talking just Windows here...

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In This Corner...

For the first time, it seems that GNU/Linux has given Microsoft something to sweat about (and not just in the eyes of fanboys)—64-bit Linux distributions were around and in use on servers long before the fumbling start of Windows XP x64 edition. Even standard excuses like "Windows has better driver support" fall flat on their face—64-bit drivers for Linux have been written by OEMs for just as long. More importantly, if you use or wish to use open source software, porting it to 64-bit Linux is as easy as recompiling it with an extra "-m64" in the command line. Of course, we're assuming that the code is written sensibly. What will happen then is that all the current 32-bit pointers will now become 64-bit pointers, allowing the program to access 64-bit memory addresses.

Windows XP x64, however, has a (debatable) edge over the Penguin. Since most of your programs will remain in their 32-bit states for a while, XP x64 includes a component called WoW (Windows on Windows, not *World of Warcraft*) that emulates a 32-bit environment for the program to run in. Even better, our informal tests showed negligible performance hits when compared to their performance under plain old 32-bit Windows.

The Penguin And Its Wings

If you really want a completely 64-bit PC, you should start window-shopping for Linux distributions. All of them are available in 64-bit versions, and differ only slightly—if at all—from their 32-bit versions (which we've tested this month).

Our Guinea Pig:
AMD Athlon64 3000+
ASUS A8N-VM motherboard with on-board sound and Ethernet
1 GB RAM
250 GB SATA hard disk
NVIDIA GeForce 6600 with 256 MB RAM

We're quite impressed with how much GNU/Linux distributions have matured in this department. For example, when we tried Ubuntu 5.04 (Breezy Badger) on this same machine a few months ago, the only device that didn't show up as "Unknown" was the processor. With their new Dapper Drake release, 64-bit Ubuntu shines—sound and Ethernet worked out-of-the-box thanks to some nifty new kernel modules, with only one annoyance: Windows XP doesn't release the Ethernet card's resources even when you shut down your PC, so to switch from Windows to Linux, you need to

The History

While 64-bit computing reached our desktops only in 2003, it's been a good 15 years since the first 64-bit processor was used—the MIPS R4000, which was used to power high-end Silicon Graphics workstations that produced movie special effects. By this time, Intel's x86 platform had established a firm death-grip on the desktop PC, supported by the accelerating popularity of Microsoft Windows. A ridiculous number of applications were written for x86 (as is true today), and to try and introduce a whole new processor architecture into the desktop market was unthinkable—who was going to write all those applications again? 64-bit computing, then, found itself in areas like special effects, high-end servers, and research.

The Fuel of The Hype....

In 1999, AMD released an extension of the x86 instruction set, called x86-64, which assumed a 64-bit processor that would run the old x86 instruction set just as easily as it would a 64-bit instruction. This meant that *everything* you could run on a regular old x86 PC would still work flawlessly, and you could also run high-performing 64-bit applications. Intel's IA-64 specification was released the same year, but it would run x86 instructions in a *compatibility mode*, which, well, are two words you don't want to say in the same breath as "high performance". AMD's Athlon64 processors hit the market in 2003, and have been showered with much love ever since.

While all of us rejoiced at the idea of moving to a 64-bit home PC with little or no hassle, much disappointment was expressed by the academic elite, who have been poking fun at x86 for years—mainly because there were far better processors than it, but without the advantage of popularity (must... resist... drawing... predictable... parallel...). Whether x86-64 was the right way to go is a battle that will rage for a while, but at least now we'll have 64-bit applications to keep us entertained while we wait.